

HLD: AI-Skill Scout – OpenClaw Agent

Goal

Develop an OpenClaw **agent** that searches GitHub for high-quality existing OpenClaw skills (SKILL.md files) relevant to developing AI models, evaluates them, and installs them safely. Invoked on demand by the user or other tools.

How It Works

1. User or tool asks: *“I need a fine-tuning skill”*
2. Agent checks if a matching skill is already installed (`openclaw skills list`)
3. If not, agent expands query via LLM (“fine-tuning” -> lora, qlora, sft, peft, unsloth...)
4. Agent searches GitHub for existing SKILL.md files
5. Agent evaluates results by repo trust + skill completeness
6. Agent presents top candidates with trust signals to user
7. User approves a skill to install
8. Agent downloads to **quarantine**, runs security scan, then installs
9. Agent logs the install and reports any remaining gaps

Pipeline (modeled on Hermes Agent)

SEARCH --> INSPECT --> QUARANTINE --> SCAN --> CONFIRM --> INSTALL --> LOG

Stage	What happens
Search	GitHub code/repo search with LLM query expansion, cached results (1hr TTL)
Inspect	Fetch SKILL.md metadata (L1: name + description only)
Quarantine	Download to temp dir, never directly to workspace
Scan	Regex-based security scan (exfiltration, injection, destructive patterns)
Confirm	Present to user with trust level + scan result. User approves.
Install	Place in workspace, verify with <code>openclaw skills check</code>
Log	Write lockfile + audit log entry for tracking

Search

Source is **GitHub** only: - `gh search code --filename SKILL.md "<query>"` – find skills by content - `gh search repos "SKILL.md <query>"` –

find skill repos - LLM query expansion for synonyms. Graceful degradation on API rate limits.

Quality Evaluation (All Free)

- **Repo trust level:** Well-known orgs (huggingface, anthropics, oracle) = HIGH, 1K+ stars = MEDIUM, community = LOW
- **Skill completeness:** Has trigger conditions, install steps, CLI commands, error handling?
- **Security scan:** 50+ regex patterns for exfiltration, injection, obfuscation, destructive commands
- **Duplication:** Does it overlap with an already-installed skill?

Trust-Level Install Policy

Trust Level	Safe patterns	Caution patterns	Dangerous patterns
HIGH (known orgs)	Auto-allow	Allow with notice	Block, ask user
MEDIUM (1K+ stars)	Auto-allow	Ask user	Block
LOW (community)	Allow	Block	Block

Decision Framework

When results come back, classify each: - **ADOPT:** Exact match, well-maintained, passes scan -> install as-is - **EXTEND:** Partial match, good foundation -> install, note limitations to user - **GAP:** Nothing suitable -> report gap, log for tracking

Constraints

- **Context pressure:** Progressive disclosure (L1 metadata -> L2 detail on selection)
- **Caching:** Search results cached with 1hr TTL to avoid redundant API calls
- **Graceful degradation:** GitHub 429/timeout -> continue with cached results
- **Install tracking:** Lockfile (installed-skills.json) + audit log per install
- **Scheduled refresh:** openclaw cron add for weekly re-scans

Detailed design: PLAN.md